Resource-Efficient and Explainable AI for Photovoltaic Anomaly Detection Under Sparse Sensing Constraints

Phase 1: Data Preprocessing and Feature Engineering Strategy

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***Abstract*—This report presents the Phase 1 data-audit, preprocessing, and feature- engineering strategy for a resource-efficient photovoltaic (PV) anomaly-detection system. The project investigates whether useful multiclass fault detection can be maintained while reducing physical sensing requirements and computational cost. The central engineering question is therefore not only whether PV operating states can be classified accurately, but also what sensing and computational configuration is sufficient under resource-constrained deployment assumptions.**

The selected PV Fault Dataset is provided by the original authors of a published photovoltaic monitoring study. After alignment of the source arrays, the raw consolidated dataset contains $N = 1,373,798$ measurement positions and $P = 7$ variables: six continuous sensor predictors and one five-class nominal target. The predictors are two DC voltage channels, two DC current channels, plane-of-array irradiance, and photovoltaic module temperature.

The Phase 1 audit found no exported NaN or infinite values. However, it identified 62 positions with simultaneous zero irradiance and temperature measurements, one slightly negative irradiance value, strong dependence between neighboring observations, substantial class imbalance, and high multicollinearity among several measured channels. Because statistically unusual sensor values may represent genuine fault behavior, the reference strategy does not automatically delete or cap outliers.

A naive random row-level split is also rejected because adjacent source positions are highly dependent. Sixteen signal-defined operating blocks are instead used to construct ordered training, validation, and test partitions, with separator regions withheld between the major partitions. All learned preprocessing quantities are restricted to training data. The report further predeclares the missingness, imputation, outlier, statistical-association, scaling, dimensionality- reduction, sensor-ablation, model-selection, and leakage-prevention strategies to be carried forward to implementation.

The resulting Phase 1 design provides a reproducible and explicitly justified preprocessing architecture for evaluating whether reduced physical sensing can preserve practically useful PV anomaly-detection performance.

***Index Terms*—Photovoltaic systems, anomaly detection, multiclass classification, data preprocessing, sensor ablation, machine learning, data leakage, explainable AI, resource-efficient AI.**

I. Introduction

Photovoltaic systems are increasingly important for decentralized and low-carbon electricity generation. Reliable electricity provision, however, depends not only on installing PV systems but also on detecting abnormal operating conditions and maintaining the systems effectively.

This motivation is particularly relevant to Sustainable Development Goal 7 (SDG 7), which emphasizes access to affordable, reliable, sustainable, and modern energy [4]. Monitoring and fault detection can contribute to this objective by supporting the reliable operation of installed PV assets.

In resource-constrained environments, monitoring architectures that depend on many sensors, continuous high-bandwidth communication, or computationally expensive models may be difficult to deploy and maintain. The project therefore investigates whether a smaller physical sensing configuration can retain useful fault-detection performance.

The central research question is:

> *What is the minimum sensing and computational configuration that can provide practically useful AI-based anomaly detection for photovoltaic systems under resource-constrained deployment assumptions?*

For Assignment 2, the first phase focuses on engineering decisions that must be made before implementation. The purpose is to audit the raw dataset, identify data-quality and statistical risks, define the preprocessing architecture, establish leakage controls, and justify planned feature-engineering choices.

The major Phase 1 decisions therefore concern data quality, missingness, imputation, outlier handling, class imbalance, feature association, multicollinearity, encoding, scaling, dimensionality reduction, data partitioning, validation, leakage prevention, and physical sensor ablation.

II. Problem Formulation and AI Task

The primary machine-learning task is **multiclass classification**.

For each observed measurement position t, the raw predictor vector is

$$\mathbf{X}_t = [vdc1_t, vdc2_t, idc1_t, idc2_t, irr_t, pvt_t]. \quad (1)$$

The variables represent two DC voltage measurements, two DC current measurements, plane-of-array irradiance, and photovoltaic module temperature.

The classifier predicts

$$\hat{y}_t = f(\mathbf{X}_t), \tag{2}$$

where

$$y_t \in \{0, 1, 2, 3, 4\}. \tag{3}$$

The five nominal classes are normal operation, short circuit, resistive degradation, open circuit, and natural building shade.

Although the target is stored using integer codes, the values are class identifiers rather than measurements. For example, class 4 is not "twice" class 2, and the numerical distance between target codes has no physical interpretation. The target is therefore never treated as a continuous or ordinal variable.

A secondary equipment-alert interpretation is also considered. Short circuit, resistive degradation, and open circuit correspond to equipment-related abnormal states. Normal operation and natural building shade are treated separately because building shade is an environmental operating condition rather than a hardware failure.

This distinction allows the project to evaluate both five-class classification performance and the operational ability to distinguish equipment-related anomalies from non-equipment conditions.

III. Dataset Selection and Provenance

The primary dataset is the *PV Fault Dataset* published by Costa [2] and associated with the monitoring system described by Lazzaretti et al. [1].

The original-author repository is:

https://github.com/clayton-h-costa/pv_fault_dataset

The repository provides two MATLAB data files:

- `dataset_elec.mat`, containing electrical measurements; and
- `dataset_amb.mat`, containing environmental and PV-temperature measurements.

The original-author source is preferred to a third-party mirror because it provides a more direct provenance chain between the data and the published monitoring study.

The measurements originate from an experimental grid-connected PV installation in Curitiba, Brazil [1]. The dataset is therefore appropriate for developing and evaluating the proposed methodology, but results from this dataset cannot establish geographic generalization to PV installations in Rwanda or other Sub-Saharan African environments.

The intended African application context motivates the project's resource-efficiency constraints. The empirical evidence used in this assignment, however, remains evidence from the documented Brazilian experimental installation.

IV. Initial Data Audit

A. Dataset Dimensions

Both source files contain the same number of aligned measurement positions:

$$N = 1,373,798. \tag{4}$$

After aligning the corresponding source positions, the consolidated raw dataset contains six predictor variables and one target variable.

Thus,

$$P_{\text{predictors}} = 6, \tag{5}$$

while the complete raw matrix contains

$$P_{\text{total}} = 7. \tag{6}$$

The conceptual raw data matrix therefore has dimensions

$$\boxed{1,373,798 \times 7}. \tag{7}$$

The two MATLAB files supply complementary variables for the same measurement positions; they are not treated as independent sets of observations.

B. Variable Inventory

Table I provides the Phase 1 variable inventory.

All six predictors are continuous physical measurements.

No discrete-count, ordinal, datetime, text, or categorical predictor variables are present in the exported dataset. The target `f_nv` is the only nominal categorical variable used in the analysis.

The target is loaded using an integer storage representation, but storage type and analytical type are distinguished: integer storage does not imply that `f_nv` is continuous or ordinal.

No timestamps, verified day identifiers, session identifiers, or system identifiers are available in the exported source arrays. Such metadata are not reconstructed or invented.

For audit and partition construction, a source-position index may be retained as metadata. This `acquisition_index` is not a physical sensor measurement and is never supplied to the classifier as a predictor.

V. Target Definition and Class Imbalance

The target variable `f_nv` contains five operating states. Table II summarizes the verified raw class distribution.

TABLE II
Raw Target Class Distribution

Code	State	Count	%
0	Normal	1,162,931	84.6508
1	Short circuit	5,999	0.4367
2	Resistive degradation	10,371	0.7549
3	Open circuit	6,024	0.4385
4	Building shade	188,473	13.7191

TABLE I
RAW DATASET VARIABLE AND MISSINGNESS INVENTORY

Variable	Description	Analytical Type	Role	Missing N	Missing %
vdc1	String 1 DC RMS voltage (V)	Continuous	Predictor	0	0.00
vdc2	String 2 DC RMS voltage (V)	Continuous	Predictor	0	0.00
idc1	String 1 DC RMS current (A)	Continuous	Predictor	0	0.00
idc2	String 2 DC RMS current (A)	Continuous	Predictor	0	0.00
irr	Plane-of-array irradiance (W/m^2)	Continuous	Predictor	0	0.00
pvt	PV module temperature (°C)	Continuous	Predictor	0	0.00
f_nv	Operating-state annotation	Nominal categorical	Target	0	0.00

The distribution is strongly imbalanced. Normal operation alone accounts for approximately 84.65% of the raw observations, whereas each of the three equipment-fault classes represents less than 1%.

Consequently, overall accuracy alone is inappropriate as the primary performance measure. A classifier predicting only the normal state would obtain approximately 84.65% raw accuracy while detecting none of the equipment faults.

The planned evaluation therefore emphasizes macro-F1, class-specific precision and recall, one-versus-rest average precision, and operational false-alarm behavior.

For class c, precision and recall are

$$Precision_c = \frac{TP_c}{TP_c + FP_c}, \tag{8}$$

and

$$Recall_c = \frac{TP_c}{TP_c + FN_c}. \tag{9}$$

The class-specific F1-score is

$$F1_c = 2\frac{Precision_c Recall_c}{Precision_c + Recall_c}. \tag{10}$$

Macro-F1 gives each class equal contribution:

$$F1_{\text{macro}} = \frac{1}{K}\sum_{c=1}^{K} F1_c, \qquad K = 5. \tag{11}$$

Class-weighted and unweighted model variants are planned for comparison. Random oversampling and SMOTE are not selected as the initial response to imbalance because duplicating or synthesizing highly dependent neighboring measurements does not create additional independent physical fault episodes.

No rebalancing is planned for validation or test data; those partitions must preserve their observed distributions for evaluation.

VI. BASELINE MISSINGNESS AND DATA QUALITY

The raw audit found no conventional missing cells in the exported variables.

Across all seven consolidated columns, the raw matrix contains

$$1,373,798 \times 7 = 9,616,586 \tag{12}$$

cells, with

$$N_{\text{missing}} = 0. \tag{13}$$

Thus, baseline exported-cell missingness is

$$\boxed{0\%}. \tag{14}$$

All 1,373,798 observed rows are complete with respect to the seven exported variables, and zero observed rows contain one or more conventional missing cells.

However, zero exported NaNs do not establish that the original monitoring process was perfectly complete. Because explicit timestamps are unavailable, intervals that were never recorded cannot be inferred reliably from the exported arrays.

The audit additionally identified 62 source positions where both irradiance and PV temperature are exactly zero. They occur in three runs of lengths 1, 32, and 29 positions. Simultaneous zeros in environmental and temperature channels may indicate an acquisition or communication event, but the available source documentation does not establish zero as a missing-value sentinel.

These 62 positions are therefore classified as *suspected quality events*, not confirmed missing observations. Their raw measurements and target labels are retained in the reference dataset, while their influence can be examined separately in a sensitivity analysis.

The audit also found no infinite values.

VII. CLEANING STRATEGY

The cleaning strategy is deliberately conservative because the project aims to detect abnormal PV behavior. An unusual measurement may be precisely the information needed to identify a fault.

The raw source data are therefore treated as immutable. Cleaning operations are applied only to derived analysis copies and are separated into physical-validity checks and statistical diagnostic screens.

A. Physical Validity Checks

The audit identified one slightly negative irradiance measurement:

$$irr = -0.0381\ W/m^2. \tag{15}$$

A physically meaningful plane-of-array irradiance magnitude should not be negative. Because this value is extremely

close to zero, the planned reference policy is to clamp this isolated small negative measurement to zero.

This rule is an engineering convention for the present analysis rather than a claim about the calibration tolerance of the original instrument.

Negative temperature values are treated differently. The minimum observed PV temperature is approximately

$$-1.65^{\circ}C, \tag{16}$$

and negative Celsius temperatures are physically possible. The audit contains 24,172 negative temperature readings, and their sign alone is not evidence of corruption.

Therefore, validity rules are defined according to the physical meaning of each sensor rather than through a generic rule such as removing all negative numerical values.

The target variable is never imputed, scaled, or physically corrected.

VIII. Missingness and Imputation Strategy

Classical missing-data analysis distinguishes three common mechanisms.

Missing Completely at Random (MCAR) means that the probability of a value being missing is independent of both observed and unobserved data.

Missing at Random (MAR) allows missingness to depend on other observed variables after conditioning on the available information.

Missing Not at Random (MNAR) describes cases in which missingness remains related to the unobserved value itself or another unobserved mechanism.

For the present exported dataset, every conventional missingness indicator is constant at zero because no NaN values are observed. Consequently, the available data provide no empirical basis for diagnosing the missingness process as MCAR, MAR, or MNAR. Assigning one of these mechanisms would therefore be unsupported.

Likewise, the 62 paired zero measurements are not automatically classified as MCAR, MAR, or MNAR because they have not been established as missing observations.

Although no baseline NaN imputation is required, a deployable monitoring pipeline must define what happens if a retained sensor becomes temporarily unavailable. The contingency strategy therefore uses a training-derived median for eligible short-duration missing predictor measurements.

For predictor x_j,

$$\tilde{x}_{j,\text{train}} = \text{median}\left\{x_{j,i} : i \in \mathcal{D}_{train}\right\}. \tag{17}$$

The median is selected rather than the mean because several sensor distributions are skewed and span substantially different low-light and generation regimes. For example, the Phase 1 training audit gives approximately:

TABLE III
Selected Training-Distribution Evidence

Sensor	Mean	Median	Skewness
idc1	2.2481	0.1105	1.0046
idc2	1.9763	0.0432	1.1450
irr	257.9285	1.8733	0.8745

The substantial mean–median differences show that a mean fallback can be strongly influenced by the mixture of operating regimes. A median fallback is more resistant to extreme values. It is nevertheless a contingency, not a claim that the training median is the physically correct value for every operating condition.

Mode imputation is not appropriate for these continuous predictors. K-nearest-neighbor imputation would add computational cost and may mix different operating or fault regimes. Interpolation and unrestricted forward filling are also avoided as the reference strategy because verified timestamps are unavailable and noncausal interpolation can implicitly use future information.

Whenever imputation is used, a per-channel availability indicator is retained so that the model can distinguish an observed value from an imputed fallback.

Short isolated unavailability and sustained sensor failure are also distinguished. Long-duration or simultaneous multi-channel loss should not be hidden by unlimited imputation; the monitoring logic may abstain when the available evidence is insufficient.

IX. Outlier Identification and Treatment

Statistical outlier detection is treated as a diagnostic tool rather than an automatic cleaning operation.

For skewed or non-normal continuous distributions, the interquartile range is

$$IQR = Q_3 - Q_1, \tag{18}$$

with screening bounds

$$L = Q_1 - 1.5(IQR) \tag{19}$$

and

$$U = Q_3 + 1.5(IQR). \tag{20}$$

For approximately symmetric continuous distributions, the standardized training-based Z-score is

$$Z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}, \tag{21}$$

with

$$|Z| > 3 \tag{22}$$

used as an extreme-value diagnostic threshold.

Any distribution-dependent bounds used later in the pipeline must be estimated from training data only.

During the Phase 1 training-partition audit, the defined global IQR and Z-score screens did not provide evidence requiring automatic treatment of any of the six measured channels. This finding does not prove that every measurement is physically correct. Rather, it provides no statistical justification for automatic winsorization or row deletion.

This distinction is especially important for fault detection. Extreme current or voltage behavior may be a genuine fault signature rather than a corrupted observation.

The reference strategy therefore applies no automatic winsorization, capping, or row deletion solely because a value is statistically unusual. Corrective treatment requires additional evidence that the measurement is an acquisition artifact or violates a defensible physical-validity rule.

X. Statistical Association Strategy

The appropriate association method depends on the analytical types of the variables being compared. The target is nominal multiclass rather than continuous, so its integer codes must not be inserted directly into ordinary numerical correlation calculations.

Table IV predeclares the association strategy.

Mutual Information (MI) is the planned primary feature-to-target association measure because it can capture nonlinear dependence and does not require the target classes to possess numerical order.

Conceptually,

$$I(X;Y) = \int \sum_{y} p(x,y) \log \left(\frac{p(x,y)}{p(x)p(y)} \right) dx, \tag{23}$$

for continuous X and discrete Y, with estimation performed numerically in implementation.

MI values are interpreted as measures of statistical association rather than causal evidence about physical fault mechanisms.

Similarly, ordinary row-wise significance tests must be interpreted cautiously because neighboring measurements are highly dependent and therefore do not represent 1.37 million independent experimental replicates.

XI. Multicollinearity Strategy

Multicollinearity is evaluated using pairwise predictor correlation and Variance Inflation Factor (VIF).

The predeclared pairwise warning threshold is

$$|r| \geq 0.85. \tag{24}$$

Pearson correlation is used for the primary linear redundancy screen, while Spearman correlation is available as a monotonic sensitivity analysis.

The Phase 1 training audit identified five predictor pairs exceeding the planned correlation warning threshold:

- `vdc1--vdc2`,
- `idc1--idc2`,
- `idc1--irr`,
- `idc1--pvt`, and
- `idc2--irr`.

VIF is defined as

$$VIF_j = \frac{1}{1 - R_j^2}, \tag{25}$$

where R_j^2 is obtained by regressing predictor j on the remaining predictors.

A VIF greater than 10 is treated as a strong multicollinearity warning, not as an automatic deletion command.

Table V reports the training-data audit values.

TABLE V
Training-Data Variance Inflation Factors

Sensor	VIF	VIF > 10
vdc1	52.168	Yes
vdc2	56.346	Yes
idc1	31.496	Yes
idc2	15.781	Yes
irr	39.293	Yes
pvt	4.070	No

Despite this strong multicollinearity, independently measured physical channels are not automatically removed.

For example, `vdc1` and `vdc2` may be highly correlated during normal operation while responding differently to a string-specific fault. Similarly, current and irradiance may be strongly associated because both respond to solar illumination, while irradiance still provides environmental context that can help distinguish changes in solar input from electrical anomalies.

The decision rule is therefore:

1) flag strongly correlated pairs and high-VIF predictors;
2) distinguish independently measured sensors from mathematically derived duplicate features;
3) retain the six physical channels in the full-sensor reference;
4) use regularization for the linear model; and
5) evaluate physical sensor removal explicitly through predefined sensor-ablation configurations.

This prevents statistical correlation alone from being mistaken for engineering redundancy.

XII. Encoding Strategy

No categorical predictor variables are present. Consequently, one-hot encoding and ordinal encoding are unnecessary for the six input features.

The target is stored using integer codes from 0 to 4, but these values represent five nominal operating states. They are retained as multiclass labels for classifier compatibility without introducing an ordinal severity interpretation.

The target is never standardized or Min–Max scaled.

If future versions of the dataset introduce nominal predictor variables, one-hot encoding with explicit unknown-category handling would be preferred for low-cardinality nominal features. Ordinal encoding would only be used when a defensible ordering actually exists.

TABLE IV
STATISTICAL ASSOCIATION STRATEGY BY VARIABLE TYPE

Variable Pair	Planned Method	Purpose / Interpretation
Continuous predictor vs. five-class nominal target	Mutual Information	Training-only nonlinear feature–target association without imposing an ordering on the target classes.
Continuous predictor vs. binary one-vs-rest class indicator	Point-biserial correlation	Descriptive class-specific association; not interpreted as causal evidence.
Continuous predictor vs. continuous predictor	Pearson correlation	Linear redundancy and multicollinearity screening.
Continuous predictor vs. continuous predictor	Spearman rank correlation	Sensitivity analysis for monotonic relationships when linearity is questionable.
Nominal predictor vs. nominal target	Chi-square, if applicable	Categorical association; not required for the current dataset because there are no categorical predictors.
Continuous predictor vs. nominal target	ANOVA-type comparison, if assumptions are defensible	Optional group-difference analysis; not used as an automatic significance claim under strongly dependent observations.

XIII. SCALING AND TRANSFORMATION STRATEGY

The six physical measurements have different units and numerical ranges: voltage is measured in volts, current in amperes, irradiance in W/m^2, and temperature in degrees Celsius.

For the Logistic Regression branch, the reference transformation is training-fitted standardization:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}. \tag{26}$$

Both μ_{train} and σ_{train} are estimated using training data only.

Standardization is appropriate for regularized linear models because different raw numerical scales would otherwise influence coefficient magnitudes and the regularization penalty disproportionately. StandardScaler does not require the original variables themselves to be Gaussian.

The Random Forest branch retains the native numerical scale. Decision trees split variables by ordered thresholds and generally do not require feature standardization.

RobustScaler is retained as a sensitivity option because it uses the median and interquartile range and is less influenced by extreme observations. It is not selected as the reference transformation because broad measurement distributions may represent legitimate PV operating regimes and fault behavior rather than contamination.

Min–Max scaling is also not selected as the reference method. Future sensor observations can fall outside the training range, and clipping such observations could conceal behavior relevant to fault detection.

An optional `log1p` transformation may be examined for nonnegative voltage, current, and irradiance measurements:

$$x' = \log(1 + x). \tag{27}$$

It is treated as a sensitivity analysis rather than a mandatory transformation.

A raw logarithm is not applied to PV temperature because Celsius values may legitimately be negative and zero degrees Celsius does not have a special multiplicative interpretation.

XIV. EXPLORATORY PCA STRATEGY

Principal Component Analysis (PCA) is planned as an exploratory analysis of the standardized six-sensor feature space.

For eigenvalue λ_j, the explained variance ratio is

$$EVR_j = \frac{\lambda_j}{\sum_k \lambda_k}. \tag{28}$$

The cumulative explained variance for the first m components is

$$CEVR_m = \sum_{j=1}^{m} EVR_j. \tag{29}$$

The implementation phase will report the smallest number of principal components required to reach 90% and 95% cumulative explained variance, with PCA fitted using training data only.

PCA is not considered evidence of physical sensor reduction.

For example,

$$PC_1 = a_1 vdc1 + a_2 vdc2 + a_3 idc1 + a_4 idc2 + a_5 irr + a_6 pvt \tag{30}$$

still requires all six underlying measurements to be acquired before the component can be computed.

Therefore,

$$\text{fewer PCA components} \not\Rightarrow \text{fewer physical sensors}. \tag{31}$$

The project's sensing-efficiency question must instead be answered by explicitly removing physical channels and measuring the resulting performance.

XV. TRAIN, VALIDATION, AND TEST STRATEGY

A conventional random row-level split is inappropriate for this dataset because neighboring source positions are highly dependent.

The Phase 1 audit found lag-one correlations above 0.9997 across the six measured channels. Adjacent rows therefore frequently represent almost identical operating conditions.

Randomly assigning neighboring observations to training and testing would allow very similar measurements to appear on both sides of the evaluation boundary and could produce overly optimistic estimates.

At the same time, explicit timestamps and verified day/session identifiers are absent. The analysis therefore does not invent calendar dates or claim that the derived groups correspond to verified days, hours, or experimental sessions.

Instead, long low-irradiance regions are used as label-independent separators between signal-defined operating regions.

A separator candidate is defined by

$$irr < 100\ W/m^2 \tag{32}$$

for at least 3,600 consecutive source positions.

The value 3,600 refers to consecutive array/source positions only. Because explicit timestamps are unavailable, it is not interpreted as 3,600 seconds or one hour.

This procedure yields 16 signal-defined operating blocks.

The planned allocation is

$$12/2/2 \tag{33}$$

blocks for training, validation, and testing:

- Blocks 1–12: training;
- Blocks 13–14: validation; and
- Blocks 15–16: final testing.

Two complete separator regions between the major partitions are withheld as evaluation buffers rather than being classified as corrupted observations.

Table VI gives the resulting partitions.

The two withheld separator regions are

$$[1006129, 1056990) \tag{34}$$

and

$$[1173967, 1227337), \tag{35}$$

containing 50,861 and 53,370 positions respectively, for a total of

$$N_{\text{separator}} = 104,231. \tag{36}$$

The final retained sample size is

$$N_{\text{retained}} = 1,269,567. \tag{37}$$

The retained-row proportions are approximately 79.2498%, 9.2139%, and 11.5363% for training, validation, and testing.

These percentages should not be confused with the 12/2/2 block allocation, which corresponds to 75%, 12.5%, and 12.5% of the 16 operating blocks. The row proportions differ because the blocks have different lengths.

All five target classes remain represented in each of the three retained major partitions.

XVI. Validation Strategy

The validation partition is designed to support model-development decisions without using the final test blocks.

Validation block 13 corresponds to

$$[1056990, 1089790) \tag{38}$$

and contains 32,800 observations. It is reserved for selecting among a small set of predeclared model settings.

Validation block 14 corresponds to

$$[1140571, 1173967) \tag{39}$$

and contains 33,396 observations. It is reserved for equipment-alert threshold selection.

Both validation subblocks contain all five operating states.

For Logistic Regression, the bounded planned search is

$$C \in \{0.1, 1.0\}, \tag{40}$$

with both unweighted and balanced class weighting.

For Random Forest, the bounded search uses 100 trees with

$$max_depth \in \{6, 10\}, \tag{41}$$

minimum leaf size 20, and unweighted or balanced class weights.

A small fixed search is preferred to a broad hyperparameter optimization because the nominal row count greatly exceeds the effective number of independent operating and fault episodes.

The selected model settings and alert threshold are intended to be frozen before final test evaluation.

XVII. Leakage Prevention Architecture

Preventing information leakage is a central requirement of the preprocessing design.

The planned architecture is

$$\begin{aligned}\text{Immutable Raw Data} &\rightarrow \text{Partition} \rightarrow \text{Fit on Training} \\ &\rightarrow \text{Transform Validation/Test.}\end{aligned} \tag{42}$$

All learned preprocessing quantities must be estimated using training data only. These include:

- imputation medians;
- scaling means and standard deviations;
- distribution-dependent transformation parameters;
- any learned feature-selection quantities;
- PCA components if PCA is evaluated;
- class-weight calculations used during model fitting; and
- fitted classifier parameters.

Validation data may be used only for the predeclared model-selection and threshold-selection roles. The final test partition must not be used to revise preprocessing decisions, hyperparameters, feature sets, or alert thresholds.

Variables excluded from the model predictor matrix include:

- the target label `f_nv`;

TABLE VI
PROPOSED ORDERED OPERATING-BLOCK PARTITION

Partition	Source Interval	Retained N	Class Counts (0/1/2/3/4)
Training	$[0, 1006129)$	1,006,129	844805 / 1226 / 5695 / 1204 / 153199
Validation	$[1056990, 1173967)$	116,977	90107 / 2412 / 2406 / 2410 / 19642
Test	$[1227337, 1373798)$	146,461	124512 / 2361 / 2270 / 2410 / 14908

- `acquisition_index`;
- derived block identifiers;
- future measurements;
- post-event information; and
- statistics fitted using validation or test observations.

This architecture ensures that learned information flows from training toward validation and testing rather than allowing held-out observations to influence model fitting.

XVIII. Sensor-Ablation Strategy

The central engineering experiment evaluates whether physical sensing requirements can be reduced while retaining useful predictive performance.

The full reference configuration is

$$F6 = \{vdc1, vdc2, idc1, idc2, irr, pvt\}. \tag{43}$$

Several physically interpretable reduced configurations are predeclared in Table VII.

TABLE VII
PLANNED SENSOR-ABLATION CONFIGURATIONS

Name	Retained Channels	Purpose
F6	vdc1, vdc2, idc1, idc2, irr, pvt	Full reference
E5	vdc1, vdc2, idc1, idc2, irr	Remove temperature
E4	vdc1, vdc2, idc1, idc2	Electrical-only
C3	idc1, idc2, irr	Current + irradiance
C2	idc1, idc2	Current-only
S4	vdc1, idc1, irr, pvt	Single-string + context
S2	vdc1, idc1	Minimal single-string

For a sparse configuration s, predictive degradation relative to the full reference is defined as

$$\Delta_s = F1_{\text{full}} - F1_{\text{sparse}}. \tag{44}$$

A small positive value represents limited degradation. A negative value would indicate that the sparse configuration outperformed the full reference on the evaluated partition.

Sensor removal is therefore evaluated empirically rather than inferred solely from correlation or PCA.

XIX. Planned Models

Three bounded model families are predeclared.

A. Engineering Baseline

A transparent engineering or rule-based baseline provides a reference for determining whether machine-learning models offer meaningful improvement over simple sensor-discrepancy logic.

B. Logistic Regression

Logistic Regression provides a lightweight and interpretable classical baseline. L2 regularization is planned to reduce coefficient instability associated with highly correlated predictors. The branch uses training-fitted standardization.

C. Random Forest

A bounded Random Forest provides a nonlinear comparison while remaining compatible with conventional laptop-based experimentation. It can model nonlinear interactions among electrical and environmental measurements without requiring feature standardization.

The comparison is designed to determine whether nonlinear interactions provide sufficient benefit to justify their additional computational cost.

XX. Predeclared Evaluation Criteria

The research question requires evaluation of both predictive quality and resource efficiency. The following project-level engineering targets are predeclared before final evaluation:

- multiclass macro-F1 ≥ 0.80;
- recall ≥ 0.90 for each equipment-fault state;
- equipment-alert false-positive rate $\leq 1\%$;
- sensor-ablation degradation $\Delta_s \leq 0.05$;
- at most four retained physical sensing channels for a configuration to qualify as sparse;
- p95 single-observation inference latency ≤ 10 ms on the reference laptop;
- serialized model artifact size ≤ 10 MiB; and
- macro-F1 degradation ≤ 0.05 under the predefined isolated 1% sensor-dropout stress test.

The criteria make the phrase "practically useful" operational for this course project and prevent success from being defined retrospectively after observing final results.

They are engineering aspirations for comparative evaluation. They are **not** claimed to be established PV safety standards, manufacturer requirements, certified maintenance thresholds, or field-validated deployment guarantees.

XXI. GitHub Architecture and Reproducibility

The project is maintained in a dedicated GitHub repository:

https://github.com/Ruganji-Prince/AISD-Assignment2-PV-Anomaly-Detection

The professor-facing Phase 1 repository is intended to serve as the primary submission artifact. Consequently, the repository documentation and this report are designed to be mutually consistent and self-contained.

The repository follows the assignment's required project architecture, including the root README and dependency specification, raw and processed data locations, Phase 1 strategy directory, Phase 2 implementation location, source-code location, and figure directory.

For the Phase 1 submission, the repository exposes the raw-data provenance, audit evidence, preprocessing strategy, split design, decision rationale, and reproducibility information required to understand the planned pipeline. Directories intended for later implementation may remain as structural placeholders without presenting completed Phase 2 evaluation results as part of the Phase 1 submission.

The root README should state explicitly:

- the PV anomaly-detection problem and research objective;
- the approved dataset source;
- $N = 1,373,798$ raw observations;
- $P = 7$ total variables, comprising six predictors and one target;
- target $Y =$ `f_nv`;
- **Task Type: Multiclass Classification**; and
- the commands and artifacts needed to inspect or reproduce the Phase 1 audit.

The repository uses `.gitignore` protections for virtual environments, caches, credentials, generated model artifacts, and processed local arrays. Any published unaltered raw-data copy is distinguished explicitly from processed or model-generated outputs.

This Phase 1 report documents audit evidence and pre-implementation engineering decisions. Completed model-evaluation outcomes, final test scores, fitted Phase 2 diagnostic results, and final sensor-ablation conclusions are outside the scope of the professor-facing Phase 1 submission.

XXII. Key Phase 1 Decisions

Table VIII summarizes the main engineering decisions and their rationale.

XXIII. Limitations

Several limitations constrain the interpretation of the planned evaluation.

First, explicit timestamps, dates, and verified session identifiers are unavailable. The sixteen operating blocks are signal-defined groups and must not be described as confirmed days, hours, or sessions.

Second, the dataset represents a single experimental PV installation in Brazil. Performance on this dataset cannot establish geographic generalization to Rwanda or other Sub-Saharan African PV systems.

Third, neighboring observations remain dependent, and the effective number of independent equipment-fault episodes is much smaller than the raw row count suggests. Large N therefore should not be interpreted as 1.37 million independent experimental trials.

Fourth, the 62 paired irradiance and temperature zeros are suspicious but cannot be confirmed as missing measurements without additional source evidence.

Fifth, high correlation among measured channels can make feature attribution unstable. Explainability results must therefore be described as model associations rather than proof of physical fault causality.

Finally, the engineering thresholds defined in this project have not been field validated as operational PV safety or maintenance standards. They serve as predeclared research criteria for comparing candidate configurations.

XXIV. Planned Phase 2 Execution

Under the Phase 1 design established in this report, the subsequent implementation phase is intended to execute the frozen preprocessing and evaluation plan using Python and Scikit-Learn [3].

The planned implementation will:

1) reproduce the fixed training, validation, and test partitions;
2) implement training-only preprocessing and leakage controls;
3) generate the required missingness, target-distribution, outlier, correlation, feature-association, transformation, and PCA diagnostic figures;
4) implement the engineering baseline, Logistic Regression, and Random Forest branches;
5) evaluate the predefined physical sensor-ablation configurations;
6) measure predictive performance, false-alarm behavior, computational latency, model size, and predefined robustness behavior;
7) compare the Phase 1 engineering assumptions with observed implementation evidence; and
8) document any justified deviations from the frozen Phase 1 strategy.

This section records the implementation plan defined at the Phase 1 checkpoint. It does not report completed Phase 2 model outcomes.

XXV. Conclusion

Phase 1 establishes a dataset-specific, leakage-aware preprocessing and feature-engineering strategy for resource-efficient photovoltaic anomaly detection.

The raw consolidated dataset contains 1,373,798 measurement positions, six continuous physical sensing channels, and one five-class nominal target. Although no conventional exported missing values are present, the audit identifies important engineering considerations, including suspicious paired zero measurements, one slightly negative irradiance value, severe class imbalance, strong adjacent-row dependence, and substantial multicollinearity among several sensors.

The resulting strategy therefore emphasizes conservative cleaning, explicit missingness contingencies, training-derived preprocessing, class-sensitive evaluation, preservation

TABLE VIII
SUMMARY OF PHASE 1 ENGINEERING DECISIONS

Issue	Decision	Reason
Dataset source	Use the original-author PV fault dataset	Provides direct provenance and labeled PV operating states.
Variable typing	Six continuous predictors; nominal multiclass target	Storage codes for `f_nv` do not imply numerical order.
Missingness mechanism	Do not assign MCAR, MAR, or MNAR without evidence	No conventional NaNs are present and suspicious zero events are not confirmed missing observations.
Missingness contingency	Use training-median fallback with availability indicators	Provides a robust fallback for eligible short sensor loss while preserving missingness information.
Negative values	Clamp only the tiny negative irradiance; retain plausible negative temperature	Cleaning follows physical interpretation rather than a generic sign rule.
Outliers	Do not automatically delete or winsorize statistical extremes	Extreme measurements may represent genuine fault behavior.
Class imbalance	Compare class weighting; avoid initial SMOTE/oversampling	Duplicating dependent rows does not create independent physical fault episodes.
Correlation	Retain all six physical channels in the full reference	Correlation does not establish engineering redundancy; ablation tests sensor removal directly.
Scaling	StandardScaler for Logistic Regression; native scale for Random Forest	Regularized linear coefficients are scale-sensitive whereas tree thresholds generally are not.
PCA	Use as exploratory mathematical dimensionality analysis	Fewer components can still require all six physical sensor inputs.
Data split	Use ordered 12/2/2 signal-defined blocks with separator buffers	Reduces leakage risk from strongly dependent neighboring observations.
Leakage prevention	Fit all learned preprocessing using training data only	Prevents held-out information from influencing fitted transformations or models.
Sensor reduction	Evaluate predefined physical sensor configurations	Directly answers the research question about minimum sensing requirements.
Success criteria	Predeclare predictive and resource targets	Prevents retrospective redefinition of "practically useful."

of potentially fault-relevant extreme measurements, algorithm-appropriate scaling, and strict isolation of held-out data.

Statistical redundancy is deliberately separated from physical sensing redundancy. Highly correlated sensors are retained in the full reference, PCA is treated as exploratory mathematical compression, and the project's minimum-sensing question is addressed through explicit physical sensor ablation.

The ordered operating-block partition and training-only preprocessing architecture define the frozen design to be carried forward to implementation. Under this Phase 1 plan, the subsequent stage evaluates whether the proposed decisions are supported empirically and whether any reduced sensing configuration can satisfy the predeclared predictive and resource-efficiency criteria.

The Phase 1 report, audit evidence, decision records, and repository documentation together provide a reproducible engineering rationale for that subsequent evaluation.

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